

Solution 1

Open sidebar

[IT Security Forensics - Project 1 - Part 1 & 2](#)

HA Forensics Walkthrough Summary40

You said:

Please, analyze the followin solution information: <https://www.hackingarticles.in/ha-forensics-vulnhub-walkthrough/>

ChatGPT said:

The "HA: Forensics" virtual machine (VM) from VulnHub is designed to provide an intermediate-level experience in cyber forensic investigation. This Capture the Flag (CTF) challenge comprises four flags that participants must uncover using various forensic techniques.

[GitHub+6vulnhub.com+6vulnhub.com+6hackingarticles.in+6nwpc-ch.org+6hackingarticles.in+6](#)

Overview of the Challenge:

Participants begin by identifying the VM's IP address using tools like netdiscover. A subsequent nmap scan reveals open services, typically SSH (port 22) and HTTP (port 80). The primary focus is on the HTTP service, which serves as the entry point for the investigation. [VIP Hacker 100+2hackingarticles.in+2GitHub+2](#)

Flag 1:

- **Enumeration:** Accessing the web page at http://<VM_IP> presents a simple interface. Using a directory brute-forcing tool like dirb uncovers an /image directory containing two images: DNA.jpg and fingerprint.jpg. [vulnhub.com+3hackingarticles.in+3GitHub+3](#)
- **Analysis:** Downloading and examining fingerprint.jpg with exiftool reveals embedded metadata, which includes the first flag. [hackingarticles.in](#)

Flag 2:

- **Further Enumeration:** Continuing with directory brute-forcing, this time targeting text files (-X .txt), exposes a tips.txt file. This file hints at the presence of a flag.zip file, which can be downloaded but is password-protected. [hackingarticles.in+1GitHub+1](#)
- **Decryption:** Within the /igolder directory, a clue.txt file is found containing a PGP private key and an encrypted message. Using an online PGP decryption tool, the message

reveals that the zip password is six characters long, starting with "for" followed by three digits.hackingarticles.in

- **Password Cracking:** A custom wordlist matching the password pattern is generated using crunch. Employing fcrackzip with this wordlist successfully cracks the zip password, yielding a PDF file that contains the second flag.[VIP Hacker 100+2hackingarticles.in+2vulnhub.com+2](https://VIPHacker100+2hackingarticles.in+2vulnhub.com+2)

Flag 3:

- **Memory Analysis:** The extracted zip also contains an lsass.DMP file. Analyzing this dump with pypykatz uncovers NTLM hashes for the user jasoos. Cracking these hashes using John the Ripper reveals the password Password@1.hackingarticles.in
- **Accessing the System:** With these credentials, an SSH session is established. Upon inspection, a Docker interface with an internal IP (172.17.0.1) is identified. Configuring Metasploit's autoroute module allows routing into this internal network.hackingarticles.in
- **Internal Enumeration:** A ping sweep identifies an FTP service on 172.17.0.2 that permits anonymous login. Within the /pub directory, a sabot001 file is found and transferred to the local machine.hackingarticles.in
- **Forensic Examination:** Using the Autopsy forensic tool to analyze sabot001 reveals a flag.txt file containing the third flag.hackingarticles.in+1vulnhub.com+1

Flag 4:

- **Privilege Escalation:** A creds.txt file, discovered during analysis, contains a Base64-encoded string. Decoding it provides the password jeenaliisagoodgirl. Switching to the forensics user with this password and checking sudo permissions (sudo -l) shows that all commands are permitted.hackingarticles.in
- **Root Access:** Executing sudo bash grants root access. Navigating to the /root directory and reading root.txt reveals the fourth and final flag.hackingarticles.in

This VM offers a comprehensive journey through various forensic investigation techniques, from metadata analysis and encrypted communications to memory dump examination and privilege escalation. It's an excellent resource for those looking to enhance their digital forensics skills.
vulnhub.com

For a visual walkthrough of capturing the first flag, you might find the following video helpful:

Sources





You said:

Does the above information correspond to the task on <https://www.vulnhub.com/entry/ha-forensics,570>?

ChatGPT said:

Yes, the information provided aligns with the "HA: Forensics" challenge available on VulnHub. This Capture the Flag (CTF) challenge is designed to offer participants hands-on experience in cyber forensic investigation, focusing on identifying and analyzing four distinct flags.

The walkthrough from Hacking Articles provides a detailed step-by-step guide to solving this challenge, covering aspects such as network scanning, service enumeration, metadata extraction, password cracking, memory dump analysis, and privilege escalation. hackingarticles.in

Both the VulnHub description and the Hacking Articles walkthrough emphasize that the primary objective is to find all four flags, rather than achieving root access. The methodologies and tools highlighted in the walkthrough are consistent with the challenge's intent to enhance participants' forensic investigation skills. vulnhub.com+2vulnhub.com+2vulnhub.com+2

Sources



You said:

The following are the Project's requirements: Download the VM from the link. Capture the flags using your Kali Linux install on your own PC Write a report analyzing your findings Include CVEs Possible Solutions to fix any problems identified Include Evidence and Chain of Custody Forms Include the steps you would take to gain evidence If you created case files, include the case file or screenshots // Can you please analyze the listed tasks one by one, assess if the given task can be scripted and craft a script for that task, based on the information available in this chat session?

ChatGPT said:

Task Analysis and Scripting Possibilities for the HA: Forensics VulnHub CTF

Let's go through each task in the project requirements, assess if the task can be **automated/scripted**, and create a script where applicable.

1. Download the VM from the link

Assessment:

- This task can be **automated** by using wget or curl in a Bash script.
- The download link may change over time, so dynamic link fetching might be needed.

Solution (Bash Script to Download the VM):

bash

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```
#!/bin/bash # Define the VulnHub download URL VM_URL="https://download.vulnhub.com/ha-forensics/HA-Forensics.ova" OUTPUT_FILE="HA-Forensics.ova" # Check if the file already exists if [ -f "$OUTPUT_FILE" ]; then echo "The file $OUTPUT_FILE already exists. Skipping download." else echo "Downloading HA: Forensics VM..." wget -O "$OUTPUT_FILE" "$VM_URL" echo "Download completed: $OUTPUT_FILE" fi
```

2. Capture the Flags Using Kali Linux

Assessment:

- This task is **partially automatable**.
- Enumeration, scanning, and certain forensic analysis steps can be scripted.
- The following actions could be automated:
 - **Network discovery** (netdiscover)
 - **Port scanning** (nmap)
 - **Web directory enumeration** (dirb)
 - **Metadata extraction** (exiftool)
 - **Brute force ZIP password cracking** (fcrackzip)
 - **Memory dump analysis** (pypykatz)

Solution (Bash Script for Automated Enumeration and Extraction in Kali Linux):

bash

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```
#!/bin/bash # Set target IP (Replace with actual discovered IP) TARGET_IP="192.168.0.XXX"
echo "Starting network discovery..." netdiscover -r 192.168.0.0/24 -P | tee netdiscover_results.txt
echo "Running Nmap scan..." nmap -sC -sV -p- -oN nmap_scan.txt $TARGET_IP echo
"Enumerating directories on the web server..." dirb http://$TARGET_IP/ -o dirb_results.txt echo
"Extracting metadata from found images..." for img in $(grep ".jpg|.png" dirb_results.txt | awk
'{print $2}'); do exiftool http://$TARGET_IP/$img >> metadata_results.txt done echo "Checking
for possible hidden files..." curl -s http://$TARGET_IP/.git/HEAD || echo "No .git repo found"
echo "Attempting to brute-force ZIP password..." fcrackzip -u -D -p rockyou.txt flag.zip | tee
zip_crack_results.txt echo "Analyzing memory dump..." pypykatz lsa minidump lsass.DMP >
memory_dump_analysis.txt echo "Script execution complete. Check output files for results."
```

3. Write a Report Analyzing Your Findings

Assessment:

- This task **cannot be fully automated** since analysis requires interpretation.
- However, a script can generate a **structured report template** based on gathered data.

Solution (Python Script to Generate Report Template):

python

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```
import datetime report = f""" # HA: Forensics VulnHub CTF - Forensic Analysis Report
**Date:** {datetime.datetime.now().strftime("%Y-%m-%d")} **Analyst:** [Your Name] ## 1.
Summary of Investigation - VM was analyzed using Kali Linux. - Tools used: nmap, netdiscover,
exiftool, dirb, pypykatz, fcrackzip. - Flags were obtained through forensic analysis. ## 2.
Identified CVEs - [List CVEs related to found vulnerabilities] ## 3. Problems Identified -
[Enumerate discovered security flaws] ## 4. Possible Solutions - [Describe mitigation
techniques] ## 5. Evidence and Chain of Custody - Evidence Collected: - [List of files with
hashes] - Chain of Custody: - [Evidence Handling Steps] ## 6. Steps Taken to Gain Evidence 1.
Network reconnaissance with nmap/netdiscover. 2. Web enumeration via dirb. 3. Metadata
extraction from images. 4. ZIP file cracking. 5. Memory dump analysis. """ with
open("Forensic_Report.txt", "w") as f: f.write(report) print("Report template generated:
Forensic_Report.txt")
```

4. Include CVEs

Assessment:

- Automatable using public CVE search APIs like [NVD API](#).
- Can be extracted from scan results.

Solution (Python Script Using NVD API to Fetch CVEs for Specific Keywords):

python

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```
import requests
def get_cves(keyword):
    url = f"https://services.nvd.nist.gov/rest/json/cves/1.0?keyword={keyword}"
    response = requests.get(url)
    if response.status_code == 200:
        cves = response.json().get("result", {}).get("CVE_Items", [])
        for cve in cves[:5]: # Limit to top 5
            print(f"CVE ID: {cve['cve']['CVE_data_meta']['ID']}")
            print(f>Description: {cve['cve']['description']['description_data'][0]['value']}\n")
# Example usage
keywords = ["PHP", "Apache", "Linux"]
for k in keywords:
    print(f"Searching CVEs for {k}...")
    get_cves(k)
```

5. Include Evidence and Chain of Custody Forms

Assessment:

- A template can be generated dynamically.

Solution (Python Script to Generate Evidence Chain Form):

python

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```
evidence_template = """ # Chain of Custody Form
**Case Number:** [Case ID]
**Investigator:** [Your Name]
**Date:** [YYYY-MM-DD]
## Evidence Collected:
1. File: [filename] - MD5: [hash]
2. File: [filename] - SHA256: [hash]
## Evidence Handling:
- Collected by: [Name]
- Stored in: [Location]
- Accessed by: [Name] on [Date]
"""
with open("Chain_of_Custody.txt", "w") as f:
    f.write(evidence_template)
print("Chain of Custody template generated.")
```

6. Include Steps to Gain Evidence

Assessment:

- A structured checklist can be generated.

Solution (Bash Script to Log Steps Automatically):

bash

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```
#!/bin/bash echo "Documenting steps taken to collect evidence..." echo "Network Discovery:
netdiscover -r 192.168.0.0/24" >> steps_log.txt echo "Port Scan: nmap -sC -sV -p- -oN
nmap_scan.txt <target>" >> steps_log.txt echo "Image Metadata Analysis: exiftool <image>" >>
steps_log.txt echo "Memory Dump Analysis: pypykatz lsa minidump lsass.DMP" >>
steps_log.txt echo "ZIP Password Crack: fcrackzip -u -D -p rockyou.txt flag.zip" >>
steps_log.txt echo "Steps logged in steps_log.txt"
```

7. Include Case Files (Screenshots)

Assessment:

- Screenshots must be taken manually, but logs and key findings can be captured automatically.

Solution (Bash Script to Auto-Capture Commands Output for Documentation):

bash

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```
#!/bin/bash mkdir -p forensic_logs nmap -sC -sV -p- <target> | tee forensic_logs/nmap_scan.txt
exiftool <image> | tee forensic_logs/metadata_analysis.txt strings flag.zip | tee
forensic_logs/zip_contents.txt echo "Evidence files saved in forensic_logs/ directory."
```

Conclusion

Most of the tasks in the HA: Forensics project can be **automated or semi-automated**. The provided scripts streamline the forensic workflow while allowing manual interpretation where needed. Let me know if you need refinements! 🚀

Search

ChatGPT can make mistakes. Check important info.

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